

z3 (5.4)

Rozwiąż równanie na przedziale $[0, 2\pi)$:

$$\sin(3x) = \sin x.$$

① Rozpisujemy $\sin(3x)$.

$$\begin{aligned}\sin(3x) &= \sin(2x + x) = \sin 2x \cos x + \cos 2x \sin x = \\ &= (2 \sin x \cos x) \cos x + (\cos^2 x - \sin^2 x) \sin x = \\ &= 2 \sin x \cos^2 x + \sin x \cos^2 x - \sin^3 x = \\ &= \sin x (2 \cos^2 x + \cos^2 x - \sin^2 x) = \\ &= \sin x (3 \cos^2 x - \sin^2 x) \\ \cos^2 x &= 1 - \sin^2 x. \\ &= \sin x (3(1 - \sin^2 x) - \sin^2 x) = \\ &= \sin x (3 - 4 \sin^2 x) = 3 \sin x - 4 \sin^3 x\end{aligned}$$

② Rozwiązujemy równanie.

$$3 \sin x - 4 \sin^3 x = \sin x$$

$$4 \sin^3 x - 2 \sin x = 0$$

$$2 \sin x (2 \sin^2 x - 1) = 0$$

$$\sin x = 0 \quad \sin^2 x = \frac{1}{2}$$

$$\sin x = \frac{\sqrt{2}}{2} \quad \sin x = -\frac{\sqrt{2}}{2}$$

$$\begin{aligned}x &= k\pi & x &= \frac{\pi}{4} + 2k\pi & x &= \frac{5\pi}{4} + 2k\pi \\ & & x &= \frac{3\pi}{4} + 2k\pi & x &= \frac{7\pi}{4} + 2k\pi, \quad k \in \mathbb{Z}\end{aligned}$$

$$\text{Dla } k=0: x=0, x=\frac{\pi}{4}, x=\frac{3\pi}{4}, x=\frac{5\pi}{4}, x=\frac{7\pi}{4}.$$

$$\text{Dla } k=1: x=\pi$$

$$\text{Odp.: } x \in \left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}.$$